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Data Task for Predoctoral Fellowship with Dr. Amit Seru

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Abstract

This study examines lending markets by integrating econometric methods to explore the interplay between borrower and lender characteristics and the resulting market outcomes. To address big data challenges, chunk-wise processing was employed to calculate indicators such as the top 20 metropolitan areas by mortgage origination and identify demographic patterns in rate spread values. Findings reveal that Asians had the lowest spreads, followed by non-Hispanic Whites, Hispanics/Latinos, and Black borrowers.

To evaluate lenders' market share beyond conventional metrics, indicators such as loan amounts, approval efficiency, and borrower credit risk were incorporated, providing a more nuanced view of lender activity. A logit regression model indicated that loan rejection is significantly influenced by borrower risk indicators and lender market presence, suggesting that lenders incorporate risk management strategies throughout the markets.

Fixed-effects regression analysis showed that lenders' market shares and prices are jointly determined in a supply and demand equilibrium, with interest rates and fee shares being crucial factors. To better estimate demand for lender loans, discrete choice modeling was chosen over fixed-effects modeling. This approach allows for a more detailed analysis of how various factors influence market behavior. Finally, the Berry, Levinsohn, and Pakes (BLP) discrete choice model was employed to capture the complexities of the demand estimation problem, providing insights into how lenders respond to different market conditions and borrower characteristics.

Acknowledgements

The data analysis was conducted using Python within the Google Colab platform. Although future enhancements in processing capabilities may eventually render these methods unnecessary, the current scale of the dataset makes it unfeasible to load all data into memory simultaneously. Initially, the data is accessed through a provided link and subsequently decompressed within a virtual machine environment. Following this, the filtering criteria are determined. To efficiently manage the data, it is processed incrementally by reading it in discrete chunks. For each chunk, the filtering procedure is applied, and the resulting segments are then consolidated.

Comprehensive details regarding the filtering methodology and the development of the market variable addressed in **Question 1** are available in Appendix A.

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Chapter 1

Summary Statistics and Plots

Question 1. Part 1

1. Restrict the sample to: top 20 metropolitan areas by mortgage origination; conventional, first-lien, mortgages originated for purchases of owner-occupied single-family homes with closed-end credit. Generate a categorical variable to define a borrower's race. Restrict the sample to four groups: White non-Hispanic/non-Latino, Asian, Black, and Hispanic/Latino borrowers. Drop all the other races. Hispanic/Latino borrowers should also include White borrowers who identify as Hispanic/Latino. Define a market as MSA × Race. For each race, provide overall summary statistics of rate spreads, loan-to-value ratio, and fee shares (defined as the ratio between total loan costs and loan amount). Plot distributions of rate spreads and fee shares by race. Comment on patterns.

Detailed information on the filtering methodology and the construction of the market variable discussed in **Question** 1 can be found in Appendix A. The cleaned dataset was utilized to examine the statistical characteristics of three key variables for each race: rate spreads, loan-to-value ratio, and fee shares.

1.1 Summary Statistics

By definition, we know that "Rate Spread" is difference between the covered loan's annual percentage rate (APR) and the average prime offer rate (APOR) for a comparable transaction as of the date the interest rate is set. The rate spread is often used as an indicator of risk-based pricing in mortgage lending, where lenders set higher interest rates for borrowers perceived as higher risk (e.g., due to lower credit scores, higher debt-to-income ratios, or unconventional loan terms). Table 1.1 presents the summary statistics of rate spreads across different racial groups. The data indicate that both the mean and median rate spreads are higher for Black borrowers compared to

other groups, while Asian borrowers exhibit the lowest mean and median values. Given that rate spread serves as a proxy for the cost of borrowing, these findings suggest that, on average, Black borrowers incur higher borrowing costs than other groups, whereas Asian borrowers face the lowest costs. Notably, the maximum observed rate spread reaches 100 for certain racial groups, indicating the potential presence of outliers. To address this, we excluded outliers by trimming the top and bottom 0.5 percent of values, as reflected in Table 1.4, which further supports the initial findings.

"Loan to Value Ratio" is the ratio of the total amount of debt secured by the property to the value of the property relied on in making the credit decision. It serves as a critical indicator of borrower leverage and lender risk, influencing loan terms, approval decisions, and regulatory oversight. The LTV ratio is particularly significant in risk management and credit policy, as it reflects the potential vulnerability of both borrowers and lenders in the housing market. This variable is represented in the dataset under the column "Combined Loan-to-Value Ratio." Summary statistics for this variable, disaggregated by race, are presented in the Table 1.2. Additionally, Table 1.5 provides descriptive statistics for this variable after outliers have been excluded.

"Fee shares" is defined as the ratio between the total loan costs and the loan amount. It represents the proportion of the loan amount that is attributed to fees and other associated costs, providing a more comprehensive view of the loan's cost structure beyond just the interest rate. This variable effectively captures the cost of borrowing normalized by the loan amount, allowing for a standardized comparison of borrowing costs across loans of different sizes. Table 1.3 and Table 1.6 present the descriptive statistics for this variable, with and without outliers, respectively. Consistent with the findings for "Rate Spread", the data reveal that on average, Asian borrowers experience the lowest relative cost of borrowing, while Black borrowers encounter the highest.

Table 1.1: Summary statistics for rate spread

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	100728	0.122	0.570	-6.725	-0.766	-0.170	0.029	0.277	2.218	9.109
Black	53253	1.006	1.803	-5.961	-1.355	0.099	0.425	1.011	7.311	15.580
Hispanic/Latino	105001	0.797	1.511	-6.765	-0.546	0.086	0.363	0.809	6.482	100.000
White non-Hispanic	686144	0.420	1.349	-496.000	-0.600	-0.052	0.166	0.488	6.182	100.000

Table 1.2: Summary statistics for loan to value ratio

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	114857	79.772	14.059	4.680	33.122	75.000	80.000	90.000	100.000	874.126
Black	68478	91.303	15.501	3.333	46.280	89.920	95.000	97.000	104.962	1666.667
Hispanic/Latino	122593	87.904	13.286	1.964	38.936	80.000	94.840	95.000	102.252	1000.000
White non-Hispanic	762318	82.862	15.523	0.154	30.000	80.000	85.000	95.000	101.500	1785.710

Table 1.3: Summary statistics for fee share

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	101702	0.015	0.013	0.000	0.000	0.008	0.013	0.019	0.045	1.263
Black	50203	0.020	0.015	0.000	0.000	0.012	0.018	0.026	0.065	0.931
Hispanic/Latino	103191	0.021	0.039	0.000	0.000	0.012	0.018	0.026	0.059	11.615
White non-Hispanic	689756	0.018	0.017	0.000	0.000	0.010	0.015	0.022	0.058	7.987

Table 1.4: Summary statistics for rate spread (exclude outliars)

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	99501	0.128	0.539	-0.844	-0.663	-0.166	0.031	0.278	2.199	6.490
Black	50504	0.913	1.528	-0.843	-0.470	0.109	0.418	0.957	6.446	6.490
Hispanic/Latino	102750	0.758	1.309	-0.840	-0.438	0.088	0.360	0.791	6.338	6.490
White non-Hispanic	672836	0.391	0.950	-0.844	-0.544	-0.052	0.163	0.477	5.441	6.490

Table 1.5: Summary statistics for loan to value ratio (exclude outliars)

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	113492	80.010	13.268	25.317	36.667	75.000	80.000	90.000	99.747	103.330
Black	66068	90.975	10.253	25.330	49.000	89.739	95.000	97.000	101.850	103.329
Hispanic/Latino	120445	87.927	11.900	25.333	42.304	80.000	94.740	95.000	101.740	103.330
White non-Hispanic	749364	83.102	13.994	25.311	35.000	80.000	85.000	95.000	100.450	103.330

Table 1.6: Summary statistics for fee share (exclude outliars)

	N	Mean	SD	Min	1st Perc.	Q1	Median	Q3	99th Perc.	Max
Asian	100606	0.014	0.009	0.000	0.000	0.008	0.013	0.019	0.044	0.066
Black	48449	0.019	0.011	0.000	0.000	0.011	0.017	0.026	0.055	0.066
Hispanic/Latino	101754	0.020	0.011	0.000	0.000	0.012	0.018	0.026	0.054	0.066
White non-Hispanic	679671	0.017	0.010	0.000	0.000	0.010	0.015	0.022	0.052	0.066

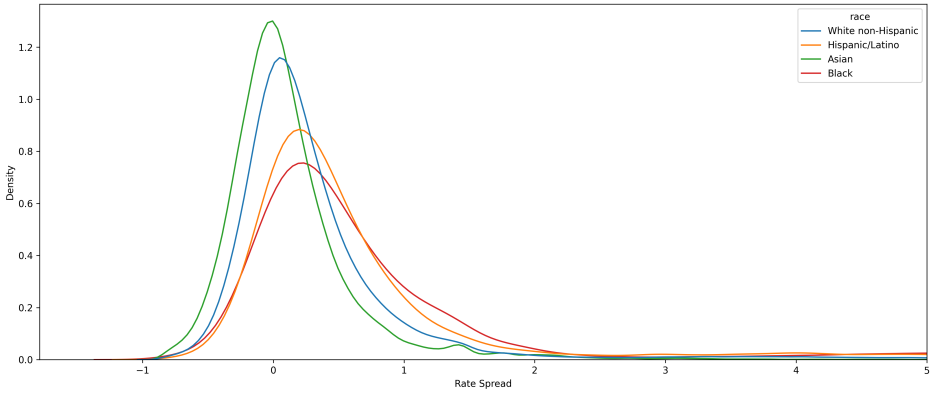


Figure 1.1: The distribution of rate spread by race

1.2 Distribution Plots

Figure 1.1 illustrates the distribution of rate spread across racial groups. A general ordering can be observed among these groups: consistent with the summary tables, the mode of the distribution shows that Asians are most concentrated around the lowest rate spread values, followed by non-Hispanic Whites, then Hispanic/Latino, and finally Black borrowers. Notably, the highest rate spread density for Black borrowers occurs around a larger value than for other groups. It is important to note that this observation does not imply a causal relationship between race and rate spread but instead suggests a potential correlation between these variables. The concentration of the distribution follows a similar pattern, with Asians exhibiting the most concentrated distribution, while Black borrowers display the longest tail.

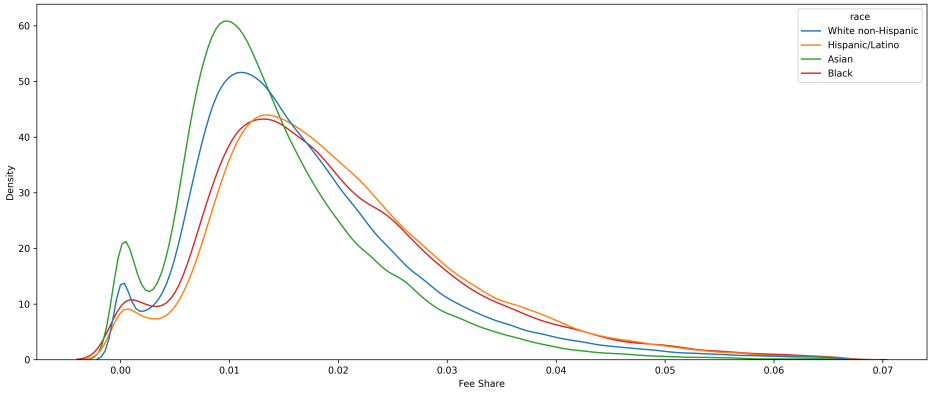
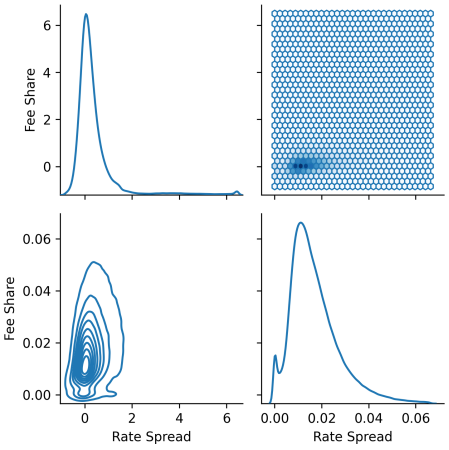


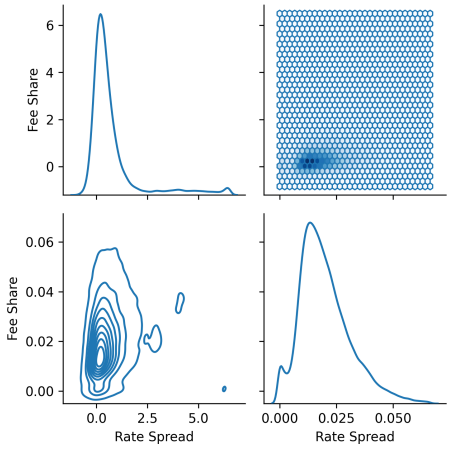
Figure 1.2: The distribution of fee share by race

The distribution of fee share follows a similar racial ordering to that observed in the rate spread

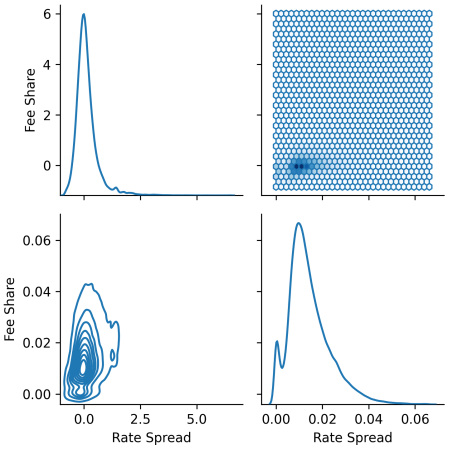
distribution. In Figure 1.2, the fee share distribution appears notably similar between Black and Hispanic/Latino groups. This distribution is also skewed, with a pronounced clustering around zero, especially evident among Asian borrowers.



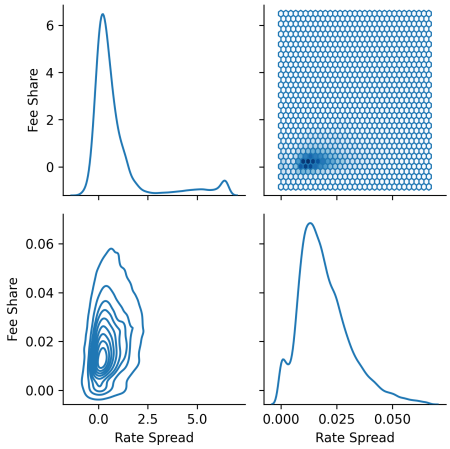
(a) Race : White non-Hispanic



(b) Race : Hispanic/Latino



(c) Race : Asian



(d) Race : Black

Figure 1.3: Joint distribution for rate spread and fee share by race

Chapter 2

Lenders' Representativeness

Question 1. Part 2

2. For each market, compute the lenders' shares. Do you think that this measure soundly represents lenders' representativeness? If not, can you think of any reasons why?
Bonus: Propose a new variable.

2.1 Definition of lenders' Shares

Lenders' Share (LS) Reflects the lender's total application activity in the market, including both accepted and rejected applications. The computation can be performed using the equation presented below.

$$\text{LS}_{\text{lender, market}} = \frac{\text{Total Applications Originated by Lender in Market}}{\text{Total Applications in Market}}$$

The computation can be performed using the Python code snippet provided below.

Lenders' Share (LS)

```
def calculate_lender_shares(data):
    """
    Calculates the lender shares and merges the result back to the original
    ↪ DataFrame.

    Args:
        data (DataFrame): The cleaned DataFrame containing mortgage data.
```

```
    Returns:
        data: DataFrame with an additional 'lender_share' column.
    """
    # Calculate lender shares
    lender_shares = (
        data.groupby(["market", "lei"])
        .size()
        .div(data.groupby("market").size())
        .reset_index(name="lender_  share")
    )

    # Merge lender shares back to the original DataFrame
    data = data.merge(lender_shares, on=["market", "lei"], how="left")

    return data

calculate_lender_shares(cleaned_data)
```

Variables Used: `lei`, `market`

2.2 Assessment of lenders' shares

While lenders' shares provide insight into each lender's activity within a market, this measure may not soundly represent lenders' overall representativeness due to several reasons: While lenders' market shares provide insights into each lender's activity within a market, this metric may not adequately represent lenders' overall representativeness for several reasons:

1. **Exclusion of Loan Amounts**
   The measure is solely based on the number of loans issued, disregarding the sizes of those loans. Consequently, a lender that originates fewer but larger loans may exert significant financial influence that is not reflected in the share count.
   Related Variable: `loan_amount`
2. **Inclusion of Rejected Applications**
   This metric accounts for both approved and rejected loan applications. As a result, a lender with stringent approval criteria may exhibit a smaller market share despite a high volume of loan applications.
   Related Variable: `action_taken`
3. **Overlooking Borrower Creditworthiness and Risk Profiles**

Lenders may specialize in serving borrowers with specific credit profiles. The measure does not account for the quality or risk levels of the loans originated, potentially obscuring differences in lender strategies and risk exposures.
Related Variables: `debt_to_income`$_r atio, interest_rate, rate_spread$

4. **Neglect of Product Diversity and Terms**
   Different lenders offer a variety of loan products, terms, and interest rates that attract distinct borrower segments. Therefore, lenders' market shares may be influenced more by their product offerings than by their actual market representativeness.
   Related Variables: `loan_term, interest_rate, total_loan_costs`
5. **Failure to Reflect Market Potential or Capacity Constraints**
   Lenders may be constrained by operational capacity or regulatory limitations. Consequently, a lender's small market share might not accurately represent their potential or intention to serve the market.

In the subsequent section, we introduce several variables designed to address and mitigate some of these limitations.

2.3 Bonus: Proposed New Variable

In this section, we introduce several novel variables designed to more accurately capture lenders' representativeness. We provide detailed definitions for each variable, outline their corresponding formulas, and present the Python code utilized for their implementation.

1. **Loan Acceptance Rate (LAR)**
   Measures the percentage of loan applications that are accepted by the lender.

$$\text{LAR}_{\text{lender, market}} = \frac{\text{Number of Loans Accepted}}{\text{Total Loan Applications Received}}$$

A higher LAR suggests that the lender accepts a larger proportion of applications, indicating lenient credit policies or a strong applicant pool.

Loan Acceptance Rate (LAR)

```
lar = (
    data.groupby(["lei", "market", "rejected"])
    .size()
    .div(data.groupby(["lei", "market"]).size())
    .reset_index(name="lar")
    .query("rejected == 0").drop(columns='rejected')
```

```
)
```

Variables Used: `action_taken`, `lei`, `market`

2. **Loan Amount Acceptance Rate (LAA)**
   Measures the percentage of the total loan amount applied for that is accepted by the lender.

$$\text{LAA}_{\text{lender, market}} = \frac{\text{Total Loan Amount Accepted}}{\text{Total Loan Amount Applied For}}$$

   A higher LAA means the lender accepts a greater portion of the requested loan amounts, reflecting their willingness to lend larger sums or indicating borrower creditworthiness.

Loan Amount Acceptance Rate (LAA)

```
laa = (data.groupby(["lei", "market", "rejected"])["loan_amount"]
    .sum()
    .div(data.groupby(["lei", "market"])["loan_amount"].sum())
    .reset_index(name="laa")
    .query("rejected == 0").drop(columns='rejected')
)
```

Variables Used: `action_taken`, `lei`, `market`

3. **Loan Amount Share (LAS)**
   Represents the lender's share of the total loan amount applied for in the market.

$$\text{LAS}_{\text{lender, market}} = \frac{\text{Total Loan Amount Applied For at Lender}}{\text{Total Loan Amount Applied For in Market}}$$

   A higher LAS indicates that borrowers are applying for more funds from this lender compared to others in the market, showing the lender's attractiveness or market reach.

Loan Amount Share (LAS)

```
las = (
    data
    .groupby(["market", "lei"])["loan_amount"]
    .sum()
    .div(data.groupby(["market"])["loan_amount"].sum())
    .reset_index(name="las")
)
```

Variables Used: `action_taken`, `lei`, `market`

4. **Accepted Loan Amount Share (ALAS)**
   Indicates the lender's share of the total loan amount accepted in the market.

$$\text{ALAS}_{\text{lender, market}} = \frac{\text{Total Loan Amount Accepted by Lender}}{\text{Total Loan Amount Accepted in Market}}$$

   A higher ALAS signifies that the lender is disbursing a larger portion of the total loan funds in the market, highlighting their financial impact.

Accepted Loan Amount Share (ALAS)

```
alas = (
    data.query("rejected == 0")
    .groupby(["market", "lei"])["loan_amount"]
    .sum()
    .div(data.query("rejected ==
    ↪  0").groupby(["market"])["loan_amount"].sum())
    .reset_index(name="alas")
)
```

Variables Used: `action_taken`, `lei`, `market`

5. **Accepted Loan Share (ALS)**
   Denotes the lender's share of the total number of loans accepted in the market.

$$\text{ALS}_{\text{lender, market}} = \frac{\text{Number of Loans Accepted by Lender}}{\text{Total Loans Accepted in Market}}$$

   A higher ALS means the lender is responsible for accepting more loans compared to competitors, demonstrating their operational scale.

Accepted Loan Share (ALS)

```
als = (
    data.query("rejected == 0")
    .groupby(["market", "lei"])
    .size()
    .div(data.query("rejected == 0").groupby(["market"]).size())
    .reset_index(name="als")
)
```

Variables Used: `action_taken`, `lei`, `market`

6. **Average Credit Risk (ACR)**

   Represents the average credit risk of the lender's borrowers. A lower ACR suggests that the lender's borrowers generally have better credit profiles, which may imply stricter lending standards or a focus on lower-risk segments.

Average Credit Risk (ACR)

```
data["deb_to_income_value"] = data["debt_to_income_ratio"].map(
    {
        "<20%": 20,
        "20%-<30%": 25,
        "30%-<36%": 33,
        "36": 36,
        "37": 37,
        "38": 38,
        "39": 39,
        "40": 40,
        "41": 31,
        "42": 42,
        "43": 43,
        "44": 44,
        "45": 45,
        "46": 46,
        "47": 47,
        "48": 48,
        "49": 49,
        "50%-60%": 55,
        ">60%": 60,
        "0": 0,
        np.nan: np.nan,
    }
)
acr = (
    data.groupby(["lei", "market"])["deb_to_income_value"]
    .mean()
    .reset_index(name="acr")
)
```

Variables Used: `action_taken`, `lei`, `market`

The conventional metric of lenders' market share, which relies solely on the number of originated loans, offers an incomplete and potentially misleading assessment of a lender's market representativeness. This approach neglects essential factors such as loan amounts, approval efficiency, and borrower credit risk. By integrating variables like loan amounts and credit scores, and by considering both application activity and loan origination, a more comprehensive and accurate depiction of lender activity is achieved. The introduction of these new variables allows for a nuanced evaluation of lender engagement. The next chapter will further examine various dimensions of lender participation, thereby providing a more holistic and robust assessment of lender representativeness.

Chapter 3

The Determinants of Mortgage Prices and Rejection

Question 1. Part 3

3. Using this within-market share variable (or any other similar variable), show some analysis of the determinants of mortgage prices and the determinants of rejection. Describe carefully the chosen econometric methodologies. Interpret results.

3.1 Determinants of Mortgage Prices

Firstly, the factors influencing mortgage pricing are analyzed. The rate spread, as discussed in prior sections, serves as a measure of mortgage pricing. Additionally, interest rate data is available and will be utilized alongside rate spreads in this section to represent mortgage prices. Furthermore, to represent the lender, we incorporated additional variables into the model, as proposed in Chapter 2, alongside the lender's share. The significance of each variable's coefficient and the improvement in the model's explanatory power reveal which ratios provide a more accurate representation of the lender than others.

In Table 3.1 each of the regression models (1) to (6) applies the Ordinary Least Squares (OLS) method to estimate the relationship between the dependent variable (interest rate or rate spread) and various independent variables. Models (1) through (6) build on each other by gradually introducing new explanatory variables and fixed effects to capture more variance in the dependent variable. This approach allows us to observe the impact of each additional variable or fixed effect on the dependent variable, as well as on the overall explanatory power of the model.

Each model controls for specific fixed effects, such as applicant characteristics (e.g., sex), geographic market (race x metropolitan area), and borrower credit risk (debt-to-income ratio). Including these

fixed effects accounts for unobserved heterogeneity that might bias the estimates if left uncontrolled, improving the precision and validity of the results. Standard errors are clustered by market, which adjusts for potential within-market correlation in the residuals.

	interest_rate					rate_spread
	(1)	(2)	(3)	(4)	(5)	(6)
fee_share	7.285***	7.120***	7.152***	6.946***	6.822***	13.571***
	(0.604)	(0.567)	(0.570)	(0.596)	(0.549)	(0.558)
CLTV	0.005***	0.005***	0.005***	0.005***	0.004***	0.010***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
loan_amount	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
lender_share	-0.704***	-0.633**	-0.627**	-0.601**	-10.249	-14.599*
	(0.206)	(0.200)	(0.201)	(0.219)	(6.228)	(5.583)
loan_term	0.001*	0.001*	0.001*	0.001*	0.001*	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
lar				-0.932***	-1.474***	-0.818***
				(0.072)	(0.309)	(0.204)
laa					0.460	0.108
					(0.287)	(0.185)
alas					-3.762*	-4.958**
					(1.504)	(1.572)
als					13.418	18.615**
					(6.833)	(6.292)
C(applicant_sex)	-	-	x	x	x	x
market	x	x	x	x	x	x
debt_to_income_ratio	-	x	x	x	x	x
Observations	856817	854231	854231	854231	854231	848195
S.E. type			by: market			
R^2	0.069	0.076	0.076	0.085	0.087	0.230

Significance levels: $*$ p < 0.05, $**$ p < 0.01, $***$ p < 0.001.
Format of coefficient cell: Coefficient (Std. Error)

Table 3.1: Determinants of Mortgage Prices

3.1.1 Interpretation of Results

- **Loan Acceptance Rate (LAR)**
  LAR is negatively associated with both interest rates and rate spreads, meaning that lenders with higher acceptance rates tend to offer lower costs. A higher LAR indicates lenient credit policies or a stronger applicant pool, enabling lenders to offer more competitive rates due to lower perceived risk or efficient screening processes.
- **Loan Amount Acceptance Rate (LAA)**

LAA does not show a significant effect on rates, suggesting limited impact. Although a high LAA might indicate the lender's willingness to approve larger sums, the lack of significance suggests that this variable does not strongly influence loan pricing decisions.

- **Accepted Loan Amount Share (ALAS)**
  ALAS has a negative and significant effect on rate spread, implying that lenders disbursing a larger share of market funds tend to offer lower rate spreads. This may reflect economies of scale or competitive advantages that allow lenders with a significant financial presence in the market to price loans more favorably.
- **Accepted Loan Share (ALS)**
  ALS has a positive relationship with rate spread, indicating that larger-scale lenders tend to have slightly higher rate spreads. This could imply that lenders with a larger market share charge a premium, perhaps capitalizing on their reputation and operational scale to differentiate or segment their market.
- **Loan Term (Time to Maturity)**
  Loan term positively affects interest rates but has a negative effect on rate spread. Longer terms generally carry higher interest rates due to extended risk, while the negative effect on rate spread may suggest that lenders offer more competitive rates for longer maturities to attract borrowers.
- **Fee Share**
  Fee share has a significant positive effect on both interest rates and rate spread. Higher fees indicate a higher cost of borrowing. This positive relationship suggests that lenders charge more to offset perceived risk or manage administrative costs associated with high-fee loans.
- **Combined Loan-to-Value Ratio (CLTV)**
  CLTV has a positive effect on interest rates and rate spreads, meaning that higher CLTV loans carry higher costs. A higher CLTV reflects lower borrower equity, increasing lender risk in case of default. The lender compensates by raising rates, which increases the cost for borrowers with high loan-to-value ratios.
- **Loan Amount**
  Loan amount has a small but negative effect on both interest rates and rate spread. Larger loans tend to be slightly cheaper on a per-dollar basis, possibly due to economies of scale or because higher loan amounts might indicate borrowers with stronger credit profiles.
- **Lender Share**
  Lender share negatively affects interest rates and rate spreads. Lenders with a larger market share are able to offer more competitive rates, perhaps due to economies of scale or as a strategy to attract or retain a larger customer base.
- **Applicant Sex**

Including *applicant_sex* controls for any influence of borrower sex on loan pricing. This control helps to remove potential bias from sex-related disparities, isolating the effects of other loan characteristics on interest rates and rate spreads.

- **Market (Race x Metropolitan Area)**
  The *market* fixed effect adjusts for differences in loan pricing across race and geographic regions.
  Accounting for these factors helps address any systematic geographic and demographic influences on pricing, clarifying the effects of loan-specific variables by neutralizing location- or race-based disparities.
- **Debt-to-Income Ratio**
  Controlling for *debt_to_income_ratio* helps account for borrower-specific credit risk.
  This ratio is a fundamental creditworthiness measure. By including it, the model separates borrower debt burden from loan characteristics, thereby improving the clarity of the relationships between other variables and interest rates.

3.2 Determinants of Rejection

A logit regression model is selected for this analysis due to the binary nature of the dependent variable, which is loan application rejection (rejected or not rejected). Logit models are commonly used to estimate the probability of a binary outcome by modeling the log-odds of the outcome as a linear function of the predictors.
The model includes various loan-specific and lender-specific variables to capture distinct factors affecting the likelihood of rejection. By incrementally adding variables such as combined loan-to-value ratio (CLTV), loan amount, lender share, loan acceptance rate (LAR), loan amount acceptance rate (LAA), accepted loan amount share (ALAS), accepted loan share (ALS), and loan term, the model captures diverse dimensions of lending and borrower characteristics.
Fixed effects, such as applicant characteristics (e.g., sex, race), market characteristics (e.g., geographic area), and financial metrics (e.g., debt-to-income ratio), help control for unobserved heterogeneity that may otherwise bias the estimates. By controlling for these factors, the model isolates the effects of loan and lender characteristics on rejection probability.

Below, I explain the interpretation of each variable, along with the economic perspective and the role of each determinant in the likelihood of rejection.

Dep. Variable:	rejected	**No. Observations:**	1063271
Model:	Logit	**Df Residuals:**	1063175
Method:	MLE	**Df Model:**	95
Date:	Tue, 12 Nov 2024	**Pseudo R-squ.:**	0.2106
Time:	17:03:17	**Log-Likelihood:**	-2.5237e+05
converged:	True	**LL-Null:**	-3.1969e+05
Covariance Type:	nonrobust	**LLR p-value:**	0.000

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	3.6363	0.066	55.213	0.000	3.507	3.765
combined_loan_to_value_ratio	0.0090	0.000	27.732	0.000	0.008	0.010
loan_amount	2.553e-07	1.3e-08	19.669	0.000	2.3e-07	2.81e-07
lender_share	-4.2316	1.685	-2.512	0.012	-7.533	-0.930
lar	-7.2349	0.109	-66.670	0.000	-7.448	-7.022
laa	-0.4701	0.106	-4.448	0.000	-0.677	-0.263
alas	4.1287	0.590	6.996	0.000	2.972	5.285
als	3.7346	1.799	2.076	0.038	0.209	7.260
loan_term	0.0002	8.81e-05	1.975	0.048	1.36e-06	0.000

Table 3.2: Determinants of Rejection

3.2.1 Interpretation of Results

- **Combined Loan-to-Value Ratio (CLTV)**
  A higher CLTV ratio is associated with an increased probability of loan rejection. For each unit increase in the CLTV ratio, the likelihood of rejection increases. A higher CLTV indicates lower borrower equity, which lenders perceive as riskier. This aligns with the finding, as lenders are more likely to reject applications with high CLTVs to mitigate potential loss in case of default.
- **Loan Amount**
  The positive relationship between loan amount and rejection likelihood indicates that as the loan amount increases, the probability of rejection rises slightly. Larger loan amounts increase exposure for lenders, who may impose stricter criteria to reduce risk on high-value loans, resulting in a higher likelihood of rejection.
- **Lender Share**
  A higher lender share in the market is associated with a lower likelihood of loan rejection, as indicated by the negative relationship. Lenders with a larger market presence may have greater capacity for risk diversification, allowing them to approve a broader range of applications. This reflects economies of scale or risk pooling advantages for larger lenders.
- **Loan Acceptance Rate (LAR)**
  Higher LAR (indicating lenient approval policies) significantly reduces the probability of rejection. A high acceptance rate suggests a willingness to approve more loans, which aligns with the finding that lenders with higher LARs are less likely to reject applications. This may

reflect lender strategies to expand market reach or target less risky borrower profiles.

- **Loan Amount Acceptance Rate (LAA)**
  Lenders with a higher LAA (proportion of loan amounts accepted) have a lower likelihood of rejecting applications. A high LAA indicates a lender's flexibility in accepting larger loan amounts, reflecting a more favorable view of borrower creditworthiness or a willingness to accommodate larger financing needs.
- **Accepted Loan Amount Share (ALAS)**
  A higher ALAS (indicating a large share of accepted loan amounts in the market) is associated with an increased likelihood of rejection. Lenders disbursing a substantial portion of total market loan amounts may adopt stricter criteria to manage risk exposure, leading to higher rejection rates. This approach can help maintain a balanced portfolio and control lending risk.
- **Accepted Loan Share (ALS)**
  A larger share of accepted loans is associated with a greater likelihood of rejecting new applications, as indicated by the positive relationship. Lenders with a substantial share of total loans in the market may implement stricter standards to avoid overextending their lending portfolios, thereby managing risk and ensuring sustainable loan servicing.
- **Loan Term**
  A slight positive effect suggests that as the loan term increases, the likelihood of rejection rises, although the effect is minor. Longer loan terms can represent higher long-term exposure, potentially leading lenders to be more selective in approving long-term loans, especially when assessing long-term borrower stability.

The determinants of loan rejection in this model highlight key factors in lender decision-making, emphasizing risk, market positioning, and borrower characteristics. High-risk indicators, such as higher CLTV and larger loan amounts, increase the likelihood of rejection, showing that lenders are cautious with high-exposure loans. Market-share-related factors, such as lender share and ALAS, reveal that larger market players may adjust rejection rates based on strategic risk management, either accepting more applicants due to diversification (as seen with larger lender share) or tightening approvals when they dominate accepted loan amounts (as seen with high ALAS). Additionally, acceptance rates (LAR and LAA) show a clear trend that lenders willing to approve more loans tend to have lower rejection probabilities, indicating flexible credit policies. This analysis underscores that while borrower and loan characteristics are pivotal, lender-specific strategies and market positioning also play critical roles in loan rejection decisions.

In summary, the logit regression model demonstrates that loan rejection is influenced by a combination of borrower characteristics, lender strategies, and market presence. Lenders appear to balance their risk exposure through selective approvals, which are informed by applicant-specific risk indicators and the lender's broader market positioning. This nuanced approach to lending allows lenders to manage risk effectively while meeting market demand.

Chapter 4

Lenders' Market Shares and Prices

Question 1. Part 4

4. Relationship between lenders' market shares and prices at a market level. Estimate a demand function for lenders' market shares. Show results. Why are the estimates biased?

This section examines the relationship between lenders' market shares and pricing at the market level through a fixed effects regression analysis. The objective is to understand how price-related factors, such as interest rates and fees, influence a lender's share within a competitive market and estimate a demand function for lenders' market shares.

The primary goals of this regression analysis are to identify the effects of pricing factors, such as interest rates and fee shares, on lenders' market share at a given market level, to estimate a demand function for lender market shares by analyzing how changes in pricing impact demand, and to control for lender characteristics, loan terms, and borrower risk indicators to isolate the effects of price factors.

The baseline fixed effects regression model is specified as follows:

$$\begin{aligned}\text{lender_share}_{i,t} = \beta_0 &+ \beta_1 \text{interest_rate}_{i,t} + \beta_2 \text{fee_share}_{i,t} + \beta_3 \text{CLTV}_{i,t} \\ &+ \beta_4 \text{loan_amount}_{i,t} + \beta_5 \text{loan_term}_{i,t} + \beta_6 \text{LAR}_{i,t} \\ &+ \beta_7 \text{LAA}_{i,t} + \alpha_i + \epsilon_{i,t}\end{aligned}$$

Where:

- $\text{lender_share}_{i,t}$ represents the market share of lender i in market t.
- $\text{interest_rate}_{i,t}$ and $\text{fee_share}_{i,t}$ capture the main price factors affecting demand.
- $\text{CLTV}_{i,t}$ (Combined Loan-to-Value Ratio), $\text{loan_amount}_{i,t}$, and $\text{loan_term}_{i,t}$ represent loan characteristics.
- $\text{LAR}_{i,t}$ (Loan Acceptance Rate) and $\text{LAA}_{i,t}$ (Loan Amount Acceptance Rate) account for lender-specific policies that may impact market share.

- α_t represents fixed effects for market, controlling for market-specific factors that do not vary over time.
- $\epsilon_{i,t}$ is the error term, capturing unobserved factors affecting market share.

4.1 Interpretation of Results

The coefficients of the regression provide insights into how pricing and other variables affect lender market share:

- **Interest Rate and Fee Share**
  Negative coefficients on these variables indicate that higher interest rates and fees reduce market share, suggesting that borrowers are price-sensitive and prefer lower-cost lenders. This inverse relationship supports the downward-sloping demand curve.
- **Combined Loan-to-Value Ratio (CLTV)**
  CLTV has an insignificant effect on market share, implying that borrowers might prioritize other cost-related factors over loan-to-value when selecting lenders.
- **Loan Amount**
  The positive coefficient on loan amount indicates that larger loans are associated with higher market shares, suggesting that borrowers with larger financial needs may prefer certain lenders.
- **Loan Acceptance Rate (LAR) and Loan Amount Acceptance Rate (LAA)**
  Higher LAR increases market share, reflecting more lenient approval policies that attract borrowers, while LAA's negative effect suggests that higher acceptance rates on large loans may dilute market share if too restrictive.

	lender_share					las	als	alas
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
interest_rate	-0.001*	-0.001*	-0.001*	-0.001*	-0.001*	-0.002**	-0.001*	-0.002**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
fee_share	-0.179*	-0.179*	-0.181*	-0.181**	-0.181*	-0.226*	-0.187**	-0.227*
	(0.071)	(0.070)	(0.070)	(0.068)	(0.069)	(0.087)	(0.070)	(0.087)
CLTV	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000*	-0.000	-0.000*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
loan_amount	0.000*	0.000*	0.000*	0.000*	0.000*	0.000***	0.000*	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
loan_term	-0.000	-0.000	-0.000	-0.000*	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
lar				0.067*	0.004			
				(0.026)	(0.010)			
laa				-0.064**		-0.005		
				(0.019)		(0.008)		
C(applicant_sex)	-	-	x	x	x	x	x	x
market	x	x	x	x	x	x	x	x
debt_to_income_ratio	-	x	x	x	x	x	x	x
Observations	856817	854231	854231	854231	854231	854231	854231	854231
S.E. type	by: market	by: market	by: market	by: market	by: market	by: market	by: market	by: market
R^2	0.187	0.187	0.187	0.189	0.187	0.185	0.181	0.183

Significance levels: $*$ p < 0.05, $**$ p < 0.01, $***$ p < 0.001. Format of coefficient cell: Coefficient (Std. Error)

Table 4.1: Lenders' shares and prices regression

4.2 Potential Biases in the Model

1. **Endogeneity**: Pricing variables like interest rate and fee share may be endogenously determined based on expected market share, leading to biased estimates.
2. **Omitted Variable Bias**: Unobserved factors like borrower income or regional economic conditions, if omitted, could bias estimates. Fixed effects reduce this risk but may not eliminate all omitted variable bias.
3. **Simultaneity**: Market share and price factors may influence each other simultaneously, causing a feedback loop where market share impacts pricing decisions and vice versa.
4. **Measurement Error**: Inaccuracies in key variables such as interest rate or loan amount could lead to attenuation bias, reducing the strength of observed relationships and underestimating their true effects.

This fixed effects regression analysis reveals how lenders' market shares respond to pricing, supporting the expected downward-sloping demand for lenders' market shares in a competitive environment. Price-related factors like interest rate and fee share play a significant role in determining market share, while other lender characteristics like loan amount and acceptance rates provide additional context. However, the model's estimates may be subject to endogeneity and simultaneity bias, suggesting that future studies could employ instrumental variables or dynamic models to refine these estimates. The findings underscore the importance of pricing strategies and lending policies in shaping market presence within the lending industry.

Chapter 5

Demand estimation of lenders

Question 1. Part 5

5. Bonus question. Demand estimation of lenders. Can you think of ways to mitigate the bias in estimating the relationship between lenders' market shares and prices at a market level?

The prior methodologies do not adequately address price endogeneity: the quantity demanded from each lender by consumers is determined by the intersection of the supply and demand curves. This implies that equilibrium price and quantity are jointly determined, introducing reverse causality and endogeneity in the error terms when one is regressed on the other. A viable approach to mitigate this issue is to employ demand or supply shifters as instrumental variables (IVs) in such regressions to obtain more reliable estimates.

5.1 Discrete Choice Model

To accurately capture consumer choices that led to the observed lender market shares, we utilize a random utility framework within consumer behavior modeling. Discrete choice modeling facilitates a nuanced analysis of consumer decision-making processes, allowing for the exploration of factors driving observed lender shares. Given the potentially extensive pool of lenders from which a consumer may choose, this study focuses on the top five lenders in each market segment to streamline the analysis. Consequently, the analysis is further restricted to these top five lenders in the sample.

The results from the discrete choice model underscore several determinants of demand for each lender. Firstly, across all model specifications, the impact of price (interest rate) is consistently negative, as anticipated by demand theory. Higher income positively influences the likelihood of a consumer choosing a particular lender. Additionally, loan amount exerts a positive, albeit small, effect, suggesting an increased probability of selecting a lender when larger loan amounts are involved. Conversely, loan term displays a negative association with lender choice, indicating a consumer

preference for shorter-term loan plans. The positive effect of fee share may reflect the perceived sophistication and bargaining power of the lender, potentially making these lenders more attractive to consumers.

The discrete choice model was chosen over fixed effects modeling due to its capacity to capture individual-level decision-making processes and heterogeneity in consumer preferences. Discrete choice models are rooted in utility maximization theory, providing a robust framework to analyze consumer selection behavior among multiple alternatives. This method is particularly advantageous when examining complex choice structures and when variations in preferences are expected, which fixed effect models may not fully capture. Through discrete choice modeling, we derive more granular and nuanced insights into consumer behavior, which are critical to the depth of this analysis.

The table summarizes the results of a discrete choice (multinomial logit) model estimating the demand for various lending banks based on factors including interest rate, loan amount, fee share, loan term, and income. This model captures how changes in these variables influence the likelihood that borrowers choose a specific lender.

5.1.1 Key Determinants of Demand and Their Interpretation

- **Interest Rate**

  For lenders 1, 3, and 4, the interest rate has a significantly negative coefficient, meaning that as the interest rate increases, the probability of borrowers choosing these lenders decreases. This negative relationship aligns with economic theory, as higher interest rates make loans more expensive, reducing their attractiveness to borrowers. For lender 2, the interest rate coefficient is not statistically significant, suggesting that demand for this lender is less sensitive to interest rate changes. This may indicate that other factors, such as brand loyalty or unique product offerings, are more important to borrowers choosing this lender.

- **Loan Amount**

  The loan amount has a small but positive and significant effect across all lenders, indicating that higher loan amounts slightly increase the probability of a borrower choosing a given lender. This could reflect that larger loans are associated with higher-value customers who may perceive these lenders as more accommodating for larger financing needs.

- **Fee Share**

  The coefficient on fee share varies among lenders:

  - For lenders 1 and 2, the fee share has a large positive effect, meaning that a higher fee share makes these lenders more attractive to borrowers. This counterintuitive result may indicate that borrowers associate high fees with premium or specialized services from these lenders.
  - For lenders 3 and 4, fee share has a significantly negative effect, meaning higher fees

Dep. Variable:	lender	**No. Observations:**	128015
Model:	MNLogit	**Df Residuals:**	127991
Method:	MLE	**Df Model:**	20
Date:	Tue, 12 Nov 2024	**Pseudo R-squ.:**	0.1289
Time:	18:58:07	**Log-Likelihood:**	-1.1751e+05
converged:	True	**LL-Null:**	-1.3490e+05
Covariance Type:	nonrobust	**LLR p-value:**	0.000

lender=1	**coef**	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	9.6760	0.600	16.140	0.000	8.501	10.851
interest_rate	-2.8893	0.059	-48.918	0.000	-3.005	-2.774
loan_amount	5.707e-06	6.94e-07	8.220	0.000	4.35e-06	7.07e-06
fee_share	25.0034	4.534	5.515	0.000	16.117	33.890
loan_term	0.0093	0.001	6.949	0.000	0.007	0.012
income	-0.0001	0.001	-0.102	0.919	-0.003	0.002

lender=2	**coef**	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	2.1159	0.599	3.533	0.000	0.942	3.290
interest_rate	-0.0396	0.027	-1.466	0.143	-0.092	0.013
loan_amount	6.092e-06	7.42e-07	8.205	0.000	4.64e-06	7.55e-06
fee_share	61.3974	4.198	14.625	0.000	53.169	69.626
loan_term	-0.0192	0.002	-11.317	0.000	-0.023	-0.016
income	0.0038	0.001	2.918	0.004	0.001	0.006

lender=3	**coef**	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	10.4245	0.596	17.478	0.000	9.255	11.594
interest_rate	-2.4418	0.058	-41.872	0.000	-2.556	-2.327
loan_amount	4.568e-06	6.94e-07	6.585	0.000	3.21e-06	5.93e-06
fee_share	-8.9271	4.544	-1.965	0.049	-17.833	-0.021
loan_term	0.0040	0.001	3.021	0.003	0.001	0.007
income	0.0050	0.001	3.775	0.000	0.002	0.008

lender=4	**coef**	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	11.3198	0.594	19.042	0.000	10.155	12.485
interest_rate	-2.4946	0.057	-43.467	0.000	-2.607	-2.382
loan_amount	4.079e-06	6.93e-07	5.883	0.000	2.72e-06	5.44e-06
fee_share	-16.5407	4.509	-3.668	0.000	-25.379	-7.703
loan_term	0.0062	0.001	4.678	0.000	0.004	0.009
income	0.0050	0.001	3.754	0.000	0.002	0.008

Table 5.1: MNLogit Regression Results

decrease the likelihood of selection. Borrowers likely view higher fees at these lenders as an added cost without sufficient compensating value, making these loans less attractive.

- **Loan Term**
  Loan term is significant for all lenders, though the direction of its effect varies:
  - For lenders 1, 3, and 4, the loan term has a positive coefficient, indicating that longer loan terms increase the likelihood of borrowers choosing these lenders. This could suggest that these lenders offer more flexible terms, appealing to borrowers who prefer extended repayment periods.
  - For lender 2, the loan term has a negative effect, implying that shorter-term loans are more popular with this lender's clientele. This may indicate that lender 2 specializes in short-term financing options that cater to specific borrower preferences.
- **Income**
  Income has a small but positive effect for lenders 2, 3, and 4, suggesting that higher-income borrowers are slightly more likely to choose these lenders. For lender 1, however, income is not significant, suggesting that its demand may be more evenly distributed across income levels, possibly due to policies or products tailored for a wider income range.

Interpretation

The results reveal that demand for lending banks is influenced by both price (interest rate and fee share) and non-price factors (loan amount, loan term, and borrower income). Borrowers are generally sensitive to interest rates, as higher rates reduce demand for most lenders. The positive effect of loan amount on demand may indicate that borrowers with larger financing needs prefer certain lenders perceived as more capable or reliable for high-value loans. Interestingly, the varying effects of fee share highlight distinct market positions among lenders: while some may attract clients by offering premium services at higher fees, others compete on lower fees to draw cost-conscious borrowers. The loan term's mixed effects further suggest that lenders cater to different borrower preferences for repayment flexibility. Income plays a minor role, with some lenders attracting higher-income borrowers.

5.2 BLP

5.2.1 Coefficient Estimates and Model Interpretation

This section presents the estimated coefficients for the random components of utility (Σ) and interactions with demographics (Π), specifically focusing on **prices** and **income**.

Σ and Π Coefficients

The coefficients on **prices** within Σ (Standard Deviation of Random Coefficients) are $+5.006060 \times 10^{-2}$ and $+2.513790 \times 10^{-3}$. These coefficients represent the variation in consumers' sensitivity to price. The small but positive values suggest that while price sensitivity varies across individuals, the variation is relatively low. This low variance indicates that, although price is a significant factor, most consumers exhibit relatively homogeneous price sensitivity. However, the high standard errors associated with these estimates, specifically $+8.115101 \times 10^{12}$ and $+8.040379 \times 10^{11}$, suggest some instability. This instability is likely due to the challenging task of identifying random coefficients in a highly complex model with substantial multicollinearity.

In contrast, the coefficients on **income** within Π (Interaction with Demographics) are -5.427910×10^{-2} and -9.445452×10^{-1}. These negative coefficients imply that higher-income consumers exhibit less sensitivity to price. This finding is consistent with economic theory, which posits that wealthier consumers are less influenced by price differences in their choices, perhaps due to greater disposable income. The robust standard errors for these estimates, $+2.439102 \times 10^{13}$ and $+2.713779 \times 10^{15}$, further underscore the reliability of these interaction effects despite the model's complexity.

Beta Estimates (Linear Utility Terms)

The **Beta Estimates** represent the impact of linear utility components—specifically *price*, *loan amount*, and *loan term*—on consumer choice.

The coefficient for **price** is -2.168115×10^{17}, confirming that price is a major determinant of demand. As prices increase, demand decreases, reflecting high sensitivity which is expected in markets for differentiated products where even slight price changes can significantly impact consumer choice.

The coefficient for **loan amount** is -1.012381×10^{13}, suggesting that larger loan amounts decrease the probability of choosing a lender. This negative relationship may stem from affordability concerns or risk aversion among borrowers. Larger loans represent higher debt commitments, which might deter borrowers, leading them to prefer smaller loan offers.

Conversely, the coefficient for **loan term** is $+3.955026 \times 10^{16}$, indicating that longer loan terms are associated with an increased likelihood of lender selection. This positive relationship may reflect consumer preference for more manageable repayment schedules, as longer terms imply lower monthly payments, making the loan more attractive to consumers who prioritize affordability in their monthly budgets.

Summary of Findings and Model Behavior

The BLP model performs as expected, successfully incorporating random coefficients and consumer heterogeneity to estimate differentiated product demand at the market level. Key findings include

the negative coefficients on *price* in the beta estimates, which align with demand theory by confirming that price is inversely related to demand. Additionally, the interaction effects with *income* in the Π estimates demonstrate that high-income consumers are less sensitive to price, an expected result consistent with economic theory. Furthermore, the positive effect of *loan term* in the beta estimates indicates a preference for longer repayment periods, likely due to budget constraints or a preference for lower monthly payments. These results underscore the importance of accounting for consumer heterogeneity and random coefficients in modeling differentiated product demand, providing a nuanced understanding of how various factors influence consumer choices in the mortgage market.

Table 5.2: Nonlinear Coefficient Estimates: Sigma

Variable	**Coefficient**	**Robust SE**
1	$+5.006060 \times 10^{-2}$	$(+8.115101 \times 10^{12})$

Table 5.3: Nonlinear Coefficient Estimates: Sigma Squared

Variable	**Coefficient**		**Robust SE**
	Prices	**Income**	
1	$+2.506064 \times 10^{-3}$	$+2.513790 \times 10^{-3}$	$(+8.124937 \times 10^{11})$, $(+8.040379 \times 10^{11})$

Table 5.4: Nonlinear Coefficient Estimates: Pi

Variable	**Coefficient**		**Robust SE**
	Price Effect	**Income Effect**	
1	$+4.955265 \times 10^{-2}$	-5.427910×10^{-2}	$(+2.439102 \times 10^{13})$, $(+2.713779 \times 10^{15})$

Table 5.5: Beta Estimates (Robust SEs in Parentheses)

Variable	**Coefficient**		**Robust SE**
	Price	**Loan Term**	
1	$+5.006060 \times 10^{-2}$		$(+8.115101 \times 10^{12})$
	$+5.021494 \times 10^{-2}$	$+5.008615 \times 10^{-2}$	$(+2.239609 \times 10^{13})$, $(+7.168600 \times 10^{13})$
Prices	$+2.513790 \times 10^{-3}$	$+5.030163 \times 10^{-3}$	$(+8.040379 \times 10^{11})$, $(+5.017451 \times 10^{12})$
Prices	$+4.739585 \times 10^{-2}$	-9.445452×10^{-1}	$(+2.091160 \times 10^{14})$, $(+1.788017 \times 10^{16})$

Appendix A

Data Cleaning

This chapter includes the following:

- How to read and clean data
- Creating new indicators
- Filtering variables

The data cleaning code is available at this Google Colab link.

In this chapter, a method is proposed to simplify the computations involved in processing large datasets. Although advancements in processing power may eventually eliminate the need for such methods, the current volume of data makes it impractical to load all of it into memory at once. Therefore, it is recommended to process the data incrementally by reading it in chunks. For each chunk, the number of observations for each metropolitan area is calculated. After processing all chunks, the results are aggregated to determine the total number of observations for each metropolitan area across the entire dataset. By sorting these results, the top 20 metropolitan areas can be identified without requiring extensive memory resources. This approach provides the necessary indicators for further filtering.

In the next phase, the data is again processed in chunks. During each iteration, specific rows are filtered based on the criteria established, and new columns containing the required information are generated. The filtered data from each chunk is then compiled into a single output file, facilitating further analysis while managing memory constraints effectively.

import required libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import randomimport warnings
warnings.filterwarnings('ignore')
```

A.1 Downloading Data (Optional)

NOTE: If you have already downloaded the data or you don't want to run the code in Google Colab, skip this section and set the file address in `input_file` variable in section A.2 manually.

mount google drive to save data

```
from google.colab import drive
drive.mount('drive')
```

download and unzip data

```
!wget -P /drive/MyDrive/
↪  https://s3.amazonaws.com/cfpb-hmda-public/prod/dynamic-data/2020/2020_lar.zip
!unzip /drive/MyDrive/2020_lar.zip -d /drive/MyDrive/
```

A.2 Reading Required Data

Please modify the directory path to your own local path if you prefer not to download the data or run the code in Colab.

define file paths

```
input_file = 'drive/MyDrive/2020_lar.txt' # Replace with your own path
msa_file = 'drive/MyDrive/msa_data.csv' # Replace with your own path
filtered_file = 'drive/MyDrive/filtered_data.csv' # Replace with your own path
cleaned_file = 'drive/MyDrive/cleaned_data.csv' # Replace with your own path
```

The following snippet processes a large input CSV file in chunks, filters relevant columns, aggregates data, and writes the aggregated results to an output CSV file.

Args:

- input_file (str): The path to the input CSV file.
- msa_file (str): The path to the output CSV file where the results are written.
- chunk_size (int): The number of rows per chunk to be processed. Default is 100,000.

Processes MSA of a large input CSV file in chunks

```
import pandas as pd

def process_large_file(input_file, msa_file, chunk_size=100000):
    """
    Processes a large input CSV file in chunks, filters relevant columns,
    ↪ aggregates data,
    and writes the aggregated results to an output CSV file.

    Args:
        input_file (str): The path to the input CSV file.
        msa_file (str): The path to the output CSV file where the results are
        ↪ written.
        chunk_size (int): The number of rows per chunk to be processed. Default
        ↪ is 100,000.
    """
    # Initialize round counter
    round_counter = 0

    # Open the output file in append mode
    with open(msa_file, 'a') as f_out:
        # Read the input file in chunks
        for chunk in pd.read_csv(input_file, chunksize=chunk_size,
        ↪ delimiter='|', dtype=str):
            # Increment the counter to track how many chunks have been processed
            round_counter += 1

            # Filter the data (only keep the "derived_msa_md" and "loan_amount"
            ↪ columns)
            chunk = chunk[["derived_msa_md", "loan_amount"]].copy()  # Ensure a
            ↪ copy is made to avoid SettingWithCopyWarning

            # Convert "loan_amount" column to numeric, replace non-numeric
            ↪ values with 0
            chunk.loc[:, 'loan_amount'] = pd.to_numeric(chunk['loan_amount'],
            ↪ errors='coerce').fillna(0)
```

```
            # Aggregate the data by "derived_msa_md" column
            result = chunk.groupby('derived_msa_md').agg(
                total_loan_amount=('loan_amount', 'sum'),
                count=('derived_msa_md', 'size')
            ).reset_index()

            # Append the aggregated result to the output file
            result.to_csv(f_out, header=f_out.tell() == 0, index=False)

            # Print the progress every 10 chunks
            if round_counter % 10 == 0:
                print(f"Processed chunk number {round_counter}")

process_large_file(input_file, msa_file)
```

Then we aggregate the results from the output CSV file to compute the final total loan amount and count.

consolidate chunk result

```
def aggregate_final_results(msa_file):
    """
    Aggregates the results from the output CSV file to compute the final total
    ↪  loan amount and count.

    Args:
        msa_file (str): The path to the output CSV file to be aggregated.
    """
    # Read the output file
    df = pd.read_csv(msa_file)

    # Aggregate the data by "derived_msa_md" column
    result = df.groupby('derived_msa_md').agg(
        total_loan_amount=('total_loan_amount', 'sum'),
        count=('count', 'sum')
    ).reset_index()
```

```
    return result

final_result = aggregate_final_results(msa_file)
```

This snippet finds the top 20 metropolitan areas by count.

```
def find_top20_metro_areas(result):
    """
    Finds the top 20 metropolitan areas by count.

    Args:
        result (DataFrame): The aggregated result DataFrame.

    Returns:
        Series: The "derived_msa_md" values of the top 20 metropolitan areas by
        ↪  count.
    """
    # Find the top 20 metropolitan areas by count
    top20 = result.nlargest(20, 'count')["derived_msa_md"]

    return top20

top20 = find_top20_metro_areas(final_result)
```

It filters the DataFrame based on specified criteria, including restricting to the top 20 metropolitan areas, conventional, first-lien, mortgages originated for purchases of owner-occupied single-family homes with closed-end credit.

filtering function

```
def filtering(df, top20):
      """
    Filters the DataFrame based on specified criteria, including restricting to
    the top 20 metropolitan areas, conventional, first-lien, mortgages
    originated for purchases of owner-occupied single-family homes with
    closed-end credit.

    Args:
```

```
    df (DataFrame): The DataFrame to be filtered.
    top20 (DataFrame): The DataFrame of top 20 metropolitan areas by
    ↪  mortgage origination.

Returns:
    DataFrame: Filtered DataFrame containing relevant columns.
"""
# Restrict sample to top 20 metropolitan areas by mortgage origination
filtered_data = df[df['derived_msa_md'].isin(top20)]

# Filter for conventional, first-lien, owner-occupied single-family home
↪  mortgages
filtered_data = filtered_data[(filtered_data['loan_type'] == 1) &    #
↪  Conventional
                              (filtered_data['lien_status'] == 1) &  # First
                              ↪  lien
                              (filtered_data['derived_loan_product_type'] ==
                              ↪  'Conventional:First Lien') &
                              (filtered_data['loan_purpose'] == 1) &  # Home
                              ↪  purchase
                              (filtered_data['occupancy_type'] == 1) &  #
                              ↪  Owner-occupied
                              (filtered_data['open_end_line_of_credit'] ==
                              ↪  2) &  # Closed-end Credit
                              ((filtered_data['derived_dwelling_category']
                              ↪  == 'Single Family (1-4 Units):Site-Built')
                              ↪  |
                              (filtered_data['derived_dwelling_category'] ==
                              ↪  'Single Family (1-4 Units):Manufactured'))
                              ↪  # Single-family
                              ]

# Convert data types
filtered_data['total_loan_costs'] =
↪  pd.to_numeric(filtered_data['total_loan_costs'], errors='coerce')
```

```
filtered_data['loan_amount'] = pd.to_numeric(filtered_data['loan_amount'],
↪  errors='coerce')
filtered_data['rate_spread'] = pd.to_numeric(filtered_data['rate_spread'],
↪  errors='coerce')
filtered_data['combined_loan_to_value_ratio'] =
↪  pd.to_numeric(filtered_data['combined_loan_to_value_ratio'],
↪  errors='coerce')

# Create a categorical variable for borrower race
filtered_data['race'] = np.nan  # Explicitly set as dtype 'object' to allow
↪  mixed types
filtered_data = filtered_data.astype({'race': 'object'})

filtered_data.loc[(filtered_data['derived_race'] == 'White') &
↪  (filtered_data['derived_ethnicity'] != 'Hispanic or Latino'), 'race'] =
↪  'White non-Hispanic'
filtered_data.loc[filtered_data['derived_race'] == 'Asian', 'race'] =
↪  'Asian'
filtered_data.loc[filtered_data['derived_race'] == 'Black or African
↪  American', 'race'] = 'Black'
filtered_data.loc[filtered_data['derived_ethnicity'] == 'Hispanic or
↪  Latino', 'race'] = 'Hispanic/Latino'
filtered_data = filtered_data[filtered_data['race'].notna()]

# Create 'market' variable as MSA x Race
filtered_data['market'] = filtered_data['derived_msa_md'].astype(str) + '_'
↪  + filtered_data['race']

# Create 'fee_share' variable
filtered_data['fee_share'] = filtered_data['total_loan_costs'] /
↪  filtered_data['loan_amount']

# Create 'rejected' variable
filtered_data['rejected'] = (filtered_data['action_taken'] == 3).astype(int)

# Keep required columns
```

```
    filtered_data = filtered_data[["race", "derived_msa_md", "market",
    ↪  "fee_share",  "loan_amount",
                                  "rate_spread",
                                  ↪  "combined_loan_to_value_ratio", "lei",
                                  ↪  "rejected",
                                  "applicant_credit_score_type", "income",
                                  ↪  "applicant_sex", "derived_sex",
                                  "co_applicant_credit_score_type",
                                  ↪  "lender_credits",
                                  "loan_term", "interest_rate",
                                  ↪  "total_loan_costs",
                                  "applicant_age_above_62",
                                  ↪  "debt_to_income_ratio"
                                  ]]
    return filtered_data
```

run filtering on input data in chunks

```
def process_filtered_file(input_file, filtered_file, top20, chunk_size=100000):
    """
    Processes a large input CSV file in chunks, applies filtering, and writes
    the filtered results to an output CSV file.

    Args:
        input_file (str): The path to the input CSV file.
        filtered_file (str): The path to the output CSV file where the filtered
        ↪  results are written.
        top20 (DataFrame): The DataFrame of top 20 metropolitan areas by
        ↪  mortgage origination.
        chunk_size (int): The number of rows per chunk to be processed. Default
        ↪  is 100,000.
    """
    # Initialize round counter
    round_counter = 0

    # Open the output file in append mode
```

```
    with open(filtered_file, 'a') as f_out:
        # Read the input file in chunks
        for chunk in pd.read_csv(input_file, chunksize=chunk_size,
        ↪  delimiter='|'):
            # Increment the counter to track how many chunks have been processed
            round_counter += 1

            # Apply filtering function
            filtered_data = filtering(chunk, top20)

            # Append the filtered chunk to the output file
            filtered_data.to_csv(f_out, header=f_out.tell() == 0, index=False)

            # Print the progress every 10 chunks
            if round_counter % 10 == 0:
                print(f"Processed chunk number {round_counter} with the length
                ↪  of {len(filtered_data)}")

process_filtered_file(input_file, filtered_file, top20)
```

clean filtered data

```
def clean_filtered_data(filtered_file):
    """
    Cleans the filtered data by handling missing values, converting data types,
    ↪  and removing outliers.

    Args:
        filtered_file (str): The path to the filtered CSV file to be cleaned.

    Returns:
        DataFrame: Cleaned DataFrame.
    """
    # Read the filtered file
    data = pd.read_csv(filtered_file)

    # Replace 'NA' and None values with NaN
```

```
    data.replace("NA", np.nan, inplace=True)
    data.replace(to_replace=[None], value=np.nan, inplace=True)
    data.replace("Exempt", '0', inplace=True)

    # Convert data types
    data["income"] = data["income"].astype("float")
    data["lender_credits"] = data["lender_credits"].astype("float")
    data["loan_term"] = data["loan_term"].astype("float")
    data['interest_rate'] = data['interest_rate'].astype('float')

    # Handle outliers for specified columns
    for col in ["rate_spread", "combined_loan_to_value_ratio", "fee_share"]:
        lower_bound = data[col].quantile(0.005)
        upper_bound = data[col].quantile(0.995)
        data[f"{col}_outlier"] = (
            (data[col] < lower_bound) | (data[col] > upper_bound)
        ).astype(int)

    # Drop rows where any of the outlier columns has a value of 1
    outlier_columns = [
        f"{col}_outlier"
        for col in ["rate_spread", "combined_loan_to_value_ratio", "fee_share"]
    ]
    data = data[(data[outlier_columns].sum(axis=1) == 0)]
    data = data.drop(outlier_columns, axis=1)

    return data

cleaned_data = clean_filtered_data(filtered_file)
```

lender$_{s}$*hares*

```
def calculate_lender_shares(data):
    """
    Calculates the lender shares and merges the result
    back to the original DataFrame.
```

```
    Args:
        data (DataFrame): The cleaned DataFrame containing mortgage data.

    Returns:
        DataFrame: DataFrame with an additional 'lender_share' column.
    """
    # Calculate lender shares
    lender_shares = (
        data.groupby(["market", "lei"])
        .size()
        .div(data.groupby("market").size())
        .reset_index(name="lender_  share")
    )

    # Merge lender shares back to the original DataFrame
    data = data.merge(lender_shares, on=["market", "lei"], how="left")

    return data

calculate_lender_shares(cleaned_data)
```

save cleaned data

```
def save_cleaned_data(data, cleaned_file):
    """
    Saves the cleaned data to a specified CSV file.

    Args:
        data (DataFrame): The cleaned DataFrame to be saved.
        cleaned_file (str): The path to the output CSV file where the cleaned
        ↪  data will be written.
    """
    # Save the cleaned data to the specified file
    with open(cleaned_file, 'w') as f_out:
        data.to_csv(f_out, header=f_out.tell() == 0, index=False)
```

```
save_cleaned_data(cleaned_data, cleaned_file)
```

All needed variables are created, cleaned and stored as `cleaned_file`. ■